COMSATS UNIVERSITY ISLAMABAD
ATTOCK CAMPUS



TOPICS IN AI

ASSIGNMENT # 01

REPORT

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Custom Reinforcement Learning Agent Implementation Using Gymnasium

Environment: FrozenLake-v1
Algorithm: Q-Learning

1. Problem Description

In this project, a Reinforcement Learning agent was developed using the FrozenLake-v1 environment from the Gymnasium library. The objective is to move an agent from the starting position to the goal while avoiding holes in a 4×4 grid.

The environment contains 16 states and four possible actions: Left, Down, Right, and Up. The agent receives a reward of +1 when it reaches the goal and 0 for other transitions. The experiment uses `is_slippery=False`, which means the agent moves in the selected direction without random slipping.

2. Methodology

Q-Learning is a model-free Reinforcement Learning algorithm. It learns the best action for each state through repeated interaction with the environment.

The Q-value update formula is:

$$Q(s,a) = Q(s,a) + \alpha\,[r + \gamma \max Q(s',a') - Q(s,a)]$$

Where:

- $Q(s,a)$ is the current value of an action in a state.
- α is the learning rate.
- r is the reward received.
- γ is the discount factor.
- s' is the next state.

An epsilon-greedy strategy was used for action selection. At the beginning, the agent explores the environment by taking random actions. As training continues, epsilon decreases, allowing the agent to use its learned Q-values more frequently.

The implementation is divided into three Python files. `main.py` handles training, `agent.py` contains the Q-Learning agent, and `utils.py` evaluates the trained

policy. The learned Q-table is saved in a NumPy file.

3. Experimental Setup and Hyperparameters

The experiment was conducted using Python, NumPy, Matplotlib, and Gymnasium.

Hyperparameter	Value
Environment	FrozenLake-v1
Number of states	16
Number of actions	4
Training episodes	15,000
Maximum steps per episode	100
Learning rate (α)	0.8
Discount factor (γ)	0.95
Initial epsilon	1.0
Minimum epsilon	0.01
Epsilon decay	0.9995
Evaluation episodes	100

4. Results and Discussion

After training for 15,000 episodes, the agent achieved a 100% test success rate over 100 evaluation episodes. The trained policy also successfully reached the goal in the saved GIF demonstration.

The training progress graph shows the average reward across groups of 100 episodes. The Q-table visualization represents the learned action values for different states and actions.

The results show that Q-Learning can learn a successful policy for this environment. However, the experiment used a deterministic environment, which is simpler than a slippery environment. The results may differ when random transitions are enabled.

A limitation of this experiment is that only one main hyperparameter configuration was evaluated. Future experiments could compare different learning rates and epsilon-decay values to study their effects on learning performance.

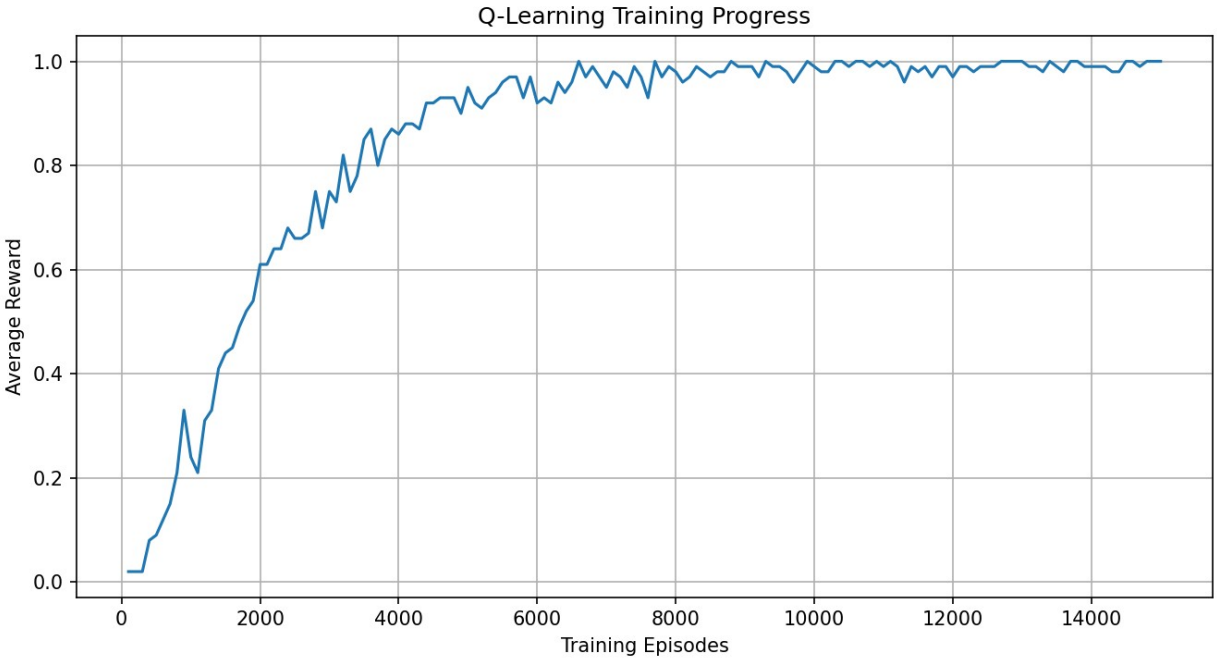


Figure 1: Training Progress Graph

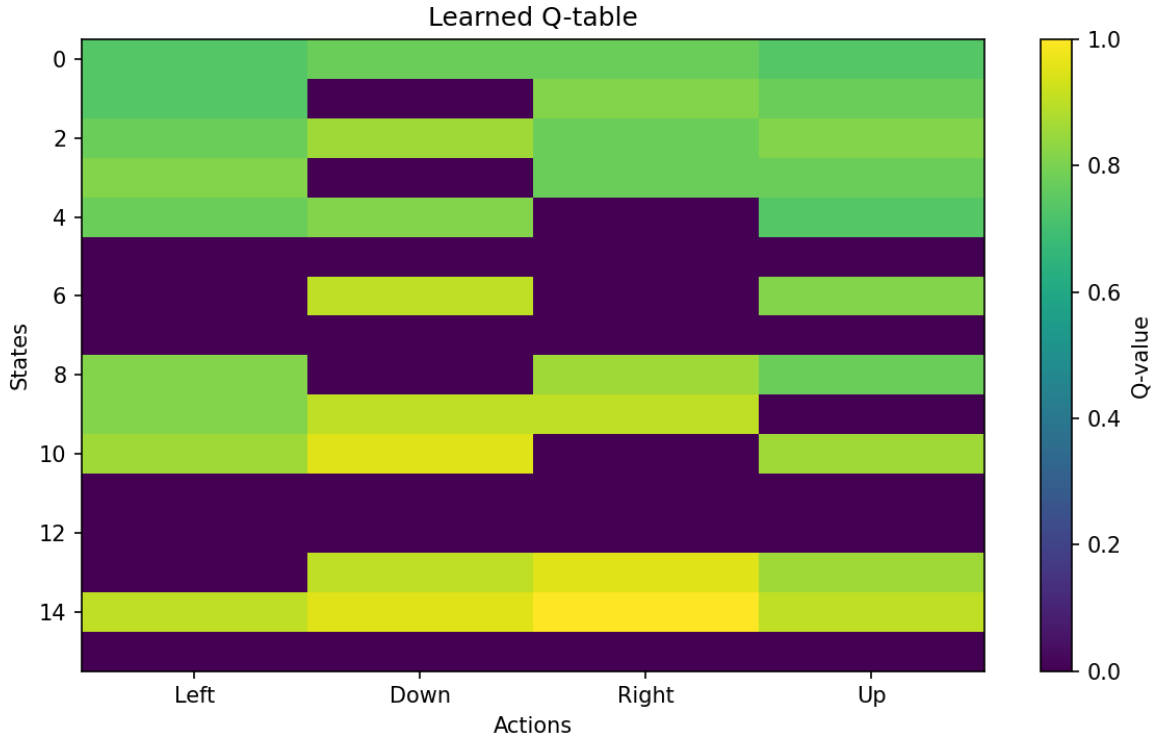


Figure 2: Q-Table Visualization

5. Conclusion

In this assignment, a Q-Learning agent was implemented and trained in the FrozenLake-v1 environment using Gymnasium. The agent learned a policy that

achieved a 100% success rate during the 100-episode evaluation. The saved Q-table and GIF demonstrate the learned behaviour.

This project provided practical experience in Reinforcement Learning, Q-value updates, exploration and exploitation, policy evaluation, and visualization of training results. Further work could evaluate the agent using different hyperparameters and stochastic environment settings